

PCB Designer Engineering Notes

RJ45 Connectors Are NOT Ethernet

Purpose

This document provides important design guidance for the PCB engineer. Several design decisions intentionally differ from what is normally assumed when RJ45 connectors appear on a printed circuit board.

Critical Design Note

IMPORTANT: The RJ45 (8P8C) connectors used throughout this project are **NOT Ethernet ports**. They are used solely as inexpensive, readily available, rugged eight-conductor connectors and cables for connecting each animatronic Derek unit to its Cluster Controller.

Signals Carried

- 5 V DC power
- Ground return
- Pan servo PWM signal
- Lift servo PWM signal
- WS2812 RGB LED data
- Optional LED data return / daisy-chain
- Reserved I/O for future expansion

Signals NOT Present

- Ethernet (10/100/1000BASE-T)
- TCP/IP
- Power over Ethernet (PoE)
- RS-485
- CAN Bus
- DMX512
- USB
- Any differential high-speed serial interface

PCB Layout Guidance

- Do NOT route differential pairs.
- Do NOT perform trace length matching.
- Do NOT control impedance for Ethernet.
- Do NOT add Ethernet magnetics.
- Do NOT include PoE circuitry.
- Optimize routing for high-current 5 V power distribution.
- Use wide power traces and large copper pours.
- Maintain solid ground return paths.

- Keep servo PWM routing clean and simple.
- Route WS2812 data with reasonable signal integrity practices.
- Separate high-current servo power from sensitive logic where practical.

Labeling Requirements

To reduce the possibility of accidental connection to network equipment, the PCB silkscreen should clearly identify the RJ45 connectors as non-Ethernet interfaces. Where space permits, include wording similar to:

WARNING – NOT ETHERNET

Animatronic Control Only

Do Not Connect to Network Equipment

Engineering Intent

The choice of RJ45 connectors is purely mechanical and economical. Standard CAT5e and CAT6 patch cables provide inexpensive, durable, widely available eight-conductor cables with locking connectors, making them ideal for field installation. The cable should always be treated as a general-purpose multi-conductor wiring harness and never as a network interface.